Data assignment: Copenhagen Perinatal Cohort

The aim of this exercise is that you should learn how to analyse and present data from a large data set. You are expected to use R for the data handling and analysis, based on what you learn from the R-tutorials.

You will be working in groups of 3-4 students. All students in a group should be active in both data analysis and writing. Each group have to make a report of around 2-3 pages.

Remember, a good report is short and concise! It should not contain commands or output from the statistical software. Make sure that the group number and the name and student code of all group members who contributed are written on the report cover page. Please send as soft copy to hfr@nexs.ku.dk before the deadline given in the lecture plan. We will go through all reports to assess if they are acceptable. You will also be given oral general feedback and we will go through the assignment in plenum.

Background:
Given the current obesity epidemic there is a lot of interest in factors predicting development of obesity. Some of these factors may be operating before birth or during infancy. It is suggested that a high growth rate before birth, expressed as a high birth weight, may be a risk factors. During infancy, short duration of breast feeding and early introduction of complementary feeding have in some studies been associated with an increased weight gain during infancy. However, it is not known if these risk factors are associated with overweight in adulthood.

The data set contains the following variables:
Baseline:

1. Sex (**SX**, 0=female, 1=male)
2. Birth weight (**BW**, in g)
3. Breastfeeding duration (**BF**, duration in weeks)
4. Complementary feeding (**CF**, time of introduction in weeks)

Follow-up at age 42 y:

5. Height (**Ht**, in cm)
6. Weight (**Wt**, in kg)

Covariate

7. Maternal marital status (**SM**, 0=married/cohabiting, 1=single mother)

Exercise

1. Describe the study population based on baseline data, ie sex, BW, BF and CF.

a) Start with **text** giving overall numbers of study participants, and descriptive statistics.
b) More detailed descriptive statistics by sex, i.e. separately for boys and girls, as well as comparisons (with p-values) are given in a **table**. This should include maternal marital status, and (infant) mean birth weight (**BW**) and the proportion with high BW (>4000 g), the proportion breastfed (**BF**) for a short time (≤8 w), and the proportion introduced to complementary feeding (**CF**) early (≤12 w).
c) Add text to highlight important estimates already shown in the table. In addition, you may present complementary information in the text, such as an estimate of the difference between boys and girls.

Remember, the accompanying text should be as short and concise as possible, and guide the reader through the table, not repeat everything.

2. Estimate a) mean BMI and b) prevalence of overweight (BMI >25 kg/m^2) in adulthood

3. Estimate the associations between exposure variables (sex, high BW, short BF, early CF) and a) BMI and b) overweight in adulthood?

4. Assess if the 4 exposure variables are independent predictors of adult BMI and overweight?

5. Examine if our covariate, single mother status, is a) a confounder, or b) an effect modifier with respect to the association between sex and adult BMI and overweight?

6. Mention strengths and limitations of the study.